

Rotorcraft Mature

The adoption of the turbine engine in the 1950s revolutionized the helicopter industry. Small, light, and powerful turbines displaced less-reliable piston engines and allowed a steep change in size, speed, and lifting capacity. Helicopters began to be used on scheduled services in the expectation that they would become more cost-effective with time.



△ Westland Dragonfly HR5 1952

Origin UK/USA

Engine 520 hp Alvis Leonides 50 radial

Top speed 105 mph (169 km/h)

The HR5, a refined, winch-equipped version of the earlier HR3, was used by the Royal Navy and tested unsuccessfully on commercial routes in England and Wales by state-owned airline BEA (British European Airways).



△ Westland Whirlwind HAR 10 1959

Origin UK

Engine 1,050 shp Bristol Siddeley Gnome turbine

Top speed 109 mph (175 km/h)

Tests with a Gnome turbine in a Whirlwind during 1959 were so encouraging that the RAF placed an order for 68 HAR 10s, and 45 older piston Whirlwinds were converted.



△ Piasecki HUP-2 Retriever 1952

Origin USA

Engine 550 hp Continental R975-46 radial

Top speed 105 mph (169 km/h)

Piasecki's HUP-2, produced for the US Navy, was the first helicopter to feature an autopilot but suffered from poor engine reliability; at least 10 HUP-2s were lost at sea.



▷ Mil Mi-6A 1957

Origin USSR

Engine 2 x 5,500 shp Soloviev D-25V turboshaft

Top speed 186 mph (299 km/h)

The Soviet Union's first turbine machine, the Mi-6 heavy transport was the largest and fastest helicopter in the world when it appeared. Its many record feats included lifting a 44,350-lb (20,117-kg) load.

▽ Mil Mi-4 1952

Origin USSR

Engine 1,675 hp Shvetsov ASh 82 14-cylinder radial

Top speed 116 mph (186 km/h)

Rushed out in response to the US deployment of helicopters in Korea, the Mi-4 has served in virtually every military and civil role from gunship to crop sprayer.



▽ NHI Sobeh H.2 Kolibri 1955

Origin Netherlands

Engine 2 x 100 hp NHI TJ5 ramjets

Top speed 100 mph (160 km/h)

Thirst for fuel conspired against the use of ramjets on rotor tips. The H.2 was trucked between revenue flights in a special "helicar" to try to make it pay.



◁ Fairey Ultra Light 1955

Origin UK

Engine Turbomeca Palouste turbojet

Top speed 98 mph (158 km/h)

Tip-jets driving the main rotor made the two-seat Ultra Light thirsty and noisy, and despite its agility—it could climb at almost 1,400 ft (427 m) per minute—it was abandoned after only six were made.

